

Final Project Report

Uncovering Deceptive Reviews Using Machine Learning and NLP Techniques with Power BI for Strategic Insights

1. Introduction

Online reviews play a critical role in influencing students' decisions when selecting educational institutions. Prospective students often rely on reviews and ratings shared on online platforms to evaluate the quality of academic programs, faculty, campus facilities, and placement opportunities. However, the increasing presence of **deceptive or fake reviews** has raised concerns regarding the reliability and credibility of online feedback.

Fake reviews are often generated to artificially improve an institution's reputation or to mislead potential students. Detecting such deceptive reviews has therefore become an important challenge for educational platforms and decision-makers.

This project aims to develop a **machine learning-based approach combined with natural language processing (NLP)** to identify potentially deceptive reviews in the education sector. Additionally, the findings are visualized using **Power BI dashboards** to provide meaningful insights and support data-driven decision-making.

2. Problem Statement

One of the main challenges in detecting fake reviews in the education sector is the **limited availability of labeled datasets specific to educational institutions**. Most publicly available fake review datasets focus on other domains such as **restaurants, hotels, e-commerce products, and services**.

Due to this limitation, building a domain-specific model becomes challenging. To address this issue, this project adopts a **cross-domain learning approach**, where fake review datasets from other domains are used to train a machine learning model. The trained model is then applied to **scraped reviews from educational institutions** to identify potentially deceptive feedback.

3. Project Objectives

The primary objectives of this project are:

1. To develop a **machine learning model** capable of detecting deceptive reviews in the education sector.
2. To apply **natural language processing (NLP) techniques** to analyze sentiment and linguistic patterns in review texts.
3. To create **interactive Power BI dashboards** that visualize review authenticity, sentiment distribution, and institutional patterns.

4. To analyze the results and generate **strategic insights and recommendations** for educational institutions.
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4. Data Collection

Data for this project was collected from two main sources.

4.1 Fake Review Dataset

Publicly available fake review datasets were collected from platforms such as **Kaggle, Yelp, and HuggingFace**. These datasets contain labeled reviews that identify whether a review is genuine or deceptive. These labeled datasets were used to train the machine learning model.

4.2 Education Reviews Dataset

Since education-specific labeled datasets are limited, reviews of educational institutions were **scraped from online review platforms**. The dataset includes:

- Institution names
- Review text
- Ratings
- Source platform

In total, **356 education-related reviews** were collected and used for analysis.

5. Data Preprocessing

Before training the model, the review text underwent several preprocessing steps to improve data quality and remove noise. These steps included:

- Converting all text to lowercase
- Removing punctuation and numbers
- Removing stopwords
- Eliminating duplicate reviews

Additionally, a filtering process was implemented to retain **education-related reviews** by identifying keywords such as *college, university, faculty, campus, placement, and students*. This helped align the training data with the target domain of the project.

6. Machine Learning Model Development

A supervised machine learning model was developed to classify reviews as **Fake** or **Genuine**.

First, the cleaned review text was transformed into numerical features using **TF-IDF (Term Frequency–Inverse Document Frequency) vectorization**. This technique converts textual data into a matrix of numerical values that represent the importance of words within the dataset.

Next, a **Logistic Regression classifier** was trained using the labeled fake review dataset. The dataset was divided into training and testing subsets to evaluate the model's performance.

Once trained, the model was applied to the scraped education reviews dataset to predict which reviews were potentially deceptive.

7. Natural Language Processing Techniques

Several NLP techniques were applied to analyze the textual characteristics of the reviews.

Text Preprocessing

Review text was cleaned and normalized through tokenization, stopword removal, and text normalization.

Feature Extraction

TF-IDF vectorization was used to represent textual data numerically, allowing machine learning algorithms to analyze patterns in the reviews.

Sentiment Analysis

Sentiment analysis was performed using the **VADER (Valence Aware Dictionary and Sentiment Reasoner)** sentiment analyzer from the NLTK library. Each review was assigned a **sentiment score** and categorized into:

- Positive
- Neutral
- Negative

This helped identify emotional patterns in the reviews.

8. Power BI Dashboard Development

An interactive dashboard was developed using **Power BI** to visualize the results of the analysis.

The dashboard includes several visualizations such as:

- **Total Reviews, Fake Reviews Detected, and Average Rating** displayed as key performance indicators.
- **Pie chart** showing the distribution of fake and genuine reviews.
- **Sentiment distribution chart** illustrating positive, negative, and neutral reviews.
- **Bar chart showing fake reviews by institution**, highlighting institutions with higher concentrations of suspicious reviews.
- **Scatter plot displaying the relationship between ratings and sentiment scores**, helping identify patterns associated with deceptive reviews.

Interactive filters were also added to allow users to explore the data by institution or sentiment category.

9. Insights from the Analysis

The analysis revealed several interesting insights.

First, the machine learning model detected **19 potentially deceptive reviews out of 356 total reviews**, representing approximately **5.3% of the dataset**. This indicates that while most reviews are genuine, a small portion may be manipulated.

Second, the sentiment analysis showed that **positive reviews dominate the dataset**, suggesting that students generally express favorable opinions about educational institutions.

Third, the institution-level analysis revealed that **some institutions have a higher concentration of predicted fake reviews**, which may indicate possible reputation management strategies.

Finally, the relationship between **ratings and sentiment scores** suggests that fake reviews often display **extremely positive sentiment and high ratings**, which can be a characteristic of promotional or exaggerated feedback.

10. Recommendations

Based on the findings of this project, several recommendations can be made.

First, educational review platforms should implement **machine learning-based fake review detection systems** to automatically identify suspicious reviews.

Second, platforms should encourage **verified student reviews** to ensure that feedback comes from authentic sources.

Third, institutions and review platforms should monitor **sentiment trends and sudden spikes in extremely positive or negative reviews**, as these patterns may indicate coordinated review activity.

Finally, review platforms can improve transparency by providing **review authenticity indicators or credibility scores** to help users evaluate the reliability of reviews.

11. Conclusion

This project demonstrates how machine learning and natural language processing techniques can be used to detect deceptive reviews in the education sector. By combining cross-domain fake review datasets with scraped education reviews, a classification model was developed to identify potentially misleading feedback.

The integration of **Power BI dashboards** further enhanced the analysis by providing clear visual insights into review authenticity, sentiment patterns, and institutional trends. These insights can help students, institutions, and online platforms make more informed decisions regarding educational reviews.

Overall, the project highlights the importance of **data-driven approaches to improving the credibility and reliability of online review systems.**
